

Scalability & Future Plan

Date	15 March 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

This document outlines the scalability strategy and future enhancement roadmap for the OptiCrop project.

S.No	Limitation	Impact	Priority
1	Limited local deployment	Restricted users	High
2	Static dataset	Reduced adaptability	Medium
3	No live weather integration	Less accurate recommendations	High

S.No	Aspect	Current State	Proposed Upgrade
1	User Load	Single server	Cloud deployment
2	Data Storage	Local files	Cloud database
3	Performance	Basic ML model	Optimized model & caching
4	Security	Basic validation	Authentication & HTTPS

Phase	Planned Enhancement	Timeline	Expected Impact
Phase 1	Improve UI/UX	1-2 Months	Better usability
Phase 2	Weather API integration	3 Months	More accurate prediction
Phase 3	Mobile application	6 Months	Wider accessibility
Phase 4	IoT sensor integration	12 Months	Real-time smart farming